MIL-M-55325(EL)

9 September 1970

MILITARY SPECIFICATION

MOUNTING MT-3802( )/ARC

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1. SCOPE

1.1 This specification covers one type of mounting upon which cipher equipment is mounted for vibration isolation, and is designated Mounting MT-3802( )/ARC. (See 6.1 and 6.5)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic, (and Connectors)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-97	Tape, Pressure Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-172	Cases, Bases, Mounting and Mounts, Vibration
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-M-13231	Marking of Electronic Equipment
MIL-F-14072	Finishes for Ground Signal Equipment

FSC 5810

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 in. x 48 in. in Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Description. - Mounting MT-3802( )/ARC consists of an upper and lower plate separated by four isolator mounts. The two plates are electrically connected through grounding straps. The cipher equipment is secured to the upper plate of the mount by the rear retainer assembly and the front self locking retainer assembly. The mounting provides vibration isolation and mounting facilities for equipment used in both rotary and fixed wing aircraft.

3.2 Construction. - The unit(s) on order shall be constructed in accordance with the following drawing and data list: DL SM-A-349414 and DL SM-A-349415

3.3 Preproduction samples. - The contractor shall furnish three (3) preproduction samples of unit(s) on order for approval, if required by the invitation for bids and contract. (See 4.3 & 6.2)

3.4 Cleaning.

3.4.1 Parts. - After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

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3.4.2 Units. - After assembly, unit(s) shall be cleaned thoroughly and shall be free from particles of foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the unit(s).

3.5 Riveting. - After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.6 Securing of parts. -

3.6.1 General. - Retainers, flanges, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.7 Finish, protective (See 4.4). - The units shall be given protective finish in accordance with Specification MIL-F-14072 and the equipment drawings. This includes finish of hardware, such as isolators, retainers, etc.

3.8 Marking (See 4.4). -

3.8.1 General. - Marking shall conform to Specification MIL-M-13231. Front panel marking shall be in accordance with the equipment drawings.

3.8.2 Serial numbers. - The following unit(s) shall be serial numbered:

Mounting MT-3802( )/ARC

3.9 Tropicalization of materials. - Materials shall be treated as follows:

3.9.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.9.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the unit(s), or during use of the finished items when used for the purpose intended.

3.9.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the unit(s) may fail to meet specified requirements when subjected to specified service conditions.

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3.9.4 Statement of treatment. - The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 3.10.3 and 6.2)

3.10 Service conditions (See 4.6). - The unit(s) shall meet the service conditions as specified in Specification MIL-E-5400 for Class 2 equipment.

3.10.1 Temperature-altitude combination (See 4.6.1). - The unit(s) shall meet the specified temperature-altitude requirements without the dummy load applied.

3.10.2 Relative humidity (See 4.6.2). - Up to 100 percent relative humidity including conditions wherein condensation takes place in the form of water.

3.10.3 Fungus (See 4.6.3). - The unit(s) shall show no evidence of viable fungus or corrosion after being subjected to 28 days of fungus test.

3.10.4 Salt spray (See 4.6.4). - The unit(s) shall be exposed, in normal operating configuration, to salt fog for 48 hours and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish.

3.10.5 Vibration (See 4.6.5). - The unit(s) shall meet the vibration Curve IV for helicopters and aircraft.

3.10.6 Shock, equipment (See 4.6.6). - The unit(s) shall meet the equipment shock test. The unit(s) shall be capable of meeting full specification performance after the shock test.

3.11 Bonding (See 4.8). - The direct current resistance shall not exceed 6 milliohms.

3.12 Sway space and snubbing (See 4.9). - The unit(s) shall meet the following requirements:

a. The unit with dummy load shall not sway more than 1 inch or snub more than 0.66 + 0.05 inch in either direction from its normal position.

b. The cushioned stop or safety device shall become operative.

c. There shall be no contact other than between snubber cushions.

d. There shall be no failure of, or damage to, the mounting base or vibration mounts.

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3.14 Interchangeability (See 4.10). - Like units, assemblies, and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable, without modification of such items or of the unit(s).

3.15 Operation inspection (See 4.12). - Each unit(s) shall be subjected to the operation inspection as specified, after applicable processes, inspection and tests are completed, to determine that each unit is operable prior to packaging.

3.16 Technical literature and spare parts. - Technical literature and spare parts shall be furnished as specified in the contract. Spare parts furnished shall be identical to corresponding parts in the unit(s) furnished or order. (See 6.2)

3.17 Workmanship. - The unit(s) shall be manufactured and assembled in accordance with the equipment drawings and applicable portions of the following requirements and paragraphs herein:

Workmanship	Requirement 9 of MIL-STD-454
Cleaning	3.4
Riveting	3.5
Securing of Parts	3.6
Tropicalization	3.9

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection:

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(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.13 and 4.14)

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B and Group C (see Tables, II and III respectively). The preproduction inspection will normally be performed in this order: (1) vibration, (2) shock, bench-handling and (3) shock, equipment, other preproduction inspection may precede, follow, or be interspersed between the foregoing.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where Required
Finish	3.7
Marking	3.8

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government -- at its discretion -- may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. - This inspection, including sampling, shall conform to Table I and the inspection procedures of Standard MIL-STD-105 using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government except that the operational inspection (4.12) shall be last. (See 6.4)

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Table I - Group A Inspection

Inspection	Req. Para.	Insp Para.	AQL Major	AQL Minor
Visual & Mechanical: MT-3802( )/ARC	3.17	4.11	1%	4%
Test, Mechanical & Electrical:				
Bonding	3.11	4.8	*1%	
Operational Inspection	3.15	4.12	0.1%	

*All electrical defects are considered major.

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table II and to the procedures for small sample inspection of MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.3 Order of inspection within Group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group B Inspection

Inspection	Req. Para.	Insp. Para.	AQL
Sway Space and Snubbing	3.12	4.9	6.5%
Interchangeability	3.14	4.10	6.5%

4.5.4 Group C inspection. - This inspection shall be as listed in Table III and shall be performed on sample unit(s) that have been subjected to and met Group A and Group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.4.1.

Table III - Group C Inspection

Inspection	Rqt Para	Insp Para
Subgroup I		
Temperature - Altitude	3.10.1	4.6.1
Humidity	3.10.2	4.6.2
Vibration	3.10.5	4.6.5

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Table III - Group C Inspection (Cont'd)

Inspection	Rqt Para.	Insp Para.
Subgroup II		
Salt Spray	3.10.4	4.6.4
Shock, Equipment	3.10.6	4.6.6
Subgroup III		
Fungus	3.10.3	4.6.3

4.5.4.1 Sampling for inspection. - Units shall be selected at random for each group of inspections in Table III as specified in 4.5.4.1.1, through 4.5.4.1.3. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. Selection of these samples shall be as specified and within 30 calendar days after selection these samples shall have been subjected to and passed the Group C inspections. (See 6.3)

4.5.4.1.1 Subgroup I. - For this subgroup, two units from the first 25 production units shall be selected for the first Group C Inspection. For subsequent Group C inspections, two units from each successive 400, or two per month, if less than 400 are produced monthly, shall be selected.

4.5.4.1.2 Subgroup II. - For this subgroup, two units (these units may be the units selected for above subgroup I) from the first 25 production units shall be selected for the first Group C Inspection. For subsequent Group C Inspection, two units from each 800, or two every three months, if less than 800 are produced during three successive months, shall be selected.

4.5.4.1.3 Subgroup III. - For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units shall be selected when one-half of the units on order have been accepted by the Government QA representative.

4.5.5 Non-compliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B

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inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.6 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

4.6 Service conditions test. - The unit(s) shall be tested in accordance with Specification MIL-T-5422, except as otherwise specified. When any service condition requires a performance check, the unit(s) shall meet the requirements of 3.15 Unit(s) shall pass a visual and mechanical inspection after the conclusion of the temperature-altitude, humidity, fungus and salt spray tests. No defects shall be allowed.

4.6.1 Temperature-altitude combination. - The unit(s) shall be tested in accordance with MIL-T-5422 Curve A to determine compliance with 3.10.1.

4.6.2 Relative humidity. - The unit(s) shall be subjected to the humidity test of MIL-T-5422 to determine compliance with 3.10.2.

4.6.3 Fungus. - The unit(s) including accessories shall be subjected to 28 days of the fungus test specified in Method 508 of MIL-STD-810. The parts and materials of the chassis shall be sprayed with the spore suspension. The unit(s) shall be subjected to the test and as specified for compliance with 3.10.3.

4.6.4 Salt spray. - The unit(s) including accessories shall be exposed in its normal operating condition to salt spray in accordance with MIL-T-5422. The unit(s) shall not be required to be operated. The unit(s) shall meet the requirement of 3.10.4.

4.6.5 Vibration. - The unit(s) shall be tested in accordance with Curve IV, MIL-T-5422 for helicopters and aircraft and meet the vibration transmissibility requirements per MIL-C-172 as modified per drawing SM-C-349413.

4.6.6 Shock, Equipment. - The unit(s) shall be subjected to the shock tests of Specification MIL-T-5422. These tests shall include the 30 G Crash Safety Test and shall meet the requirements of 3.10.6.

4.7 Bonding test. - The d-c resistance shall be measured with an instrument having an accuracy of $\pm$1.0 percent. The unit(s) with the bonding

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straps in place shall be mounted on a flat, bare-metal plate in a manner comparable to actual aircraft installation. The resistance shall be measured between the holding devices and the metal plate. The requirements of 3.11 shall be met.

4.8 Sway space and snubbing test. - The unit(s), with minimum rated load applied, shall be subjected to a force extending through the center of gravity of the unit(s), along each of its three principal axes, in turn. This force shall be seven and one-half times the maximum weight of the unit(s). At the conclusion of each application of force there shall be no damage to the unit(s) and the requirements of 3.12 shall be met.

4.9 Inspection for dimensional interchangeability. - All mechanical dimensions specified on the drawings shall be measured to determine compliance with physical interchangeability requirement of 3.14. When a dimension is not within specified or design limits, it shall be considered a major defect.

4.10 Visual and mechanical inspection. - Workmanship shall be inspected in accordance with Standard MIL-STD-252.

4.11 Operational inspection. - Depress the upper plate of the unit with a uniform force (18 to 24 pounds) the isolators shall compress. Remove the uniform force, the upper plate shall return to its no load position and be parallel with the bottom plate. The requirements of 3.15 shall be met.

4.12 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the packaging requirements of Section 5.

4.13 Rough handling test (preparation for delivery). - When rough handling test in accordance with Specification MIL-P-116 is required by the contract the operational inspection test 4.12, shall be conducted to determine freedom from operational malfunction caused by rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be Level A or C as specified. (See 6.1)

5.1.1 Level A. -

5.1.1.1 Cleaning. - Each Mounting MT-3802( )/ARC shall be cleaned in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. - Each Mounting MT-3802( )/ARC shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. - Each technical literature shall be packaged method IC-1.

5.1.1.4.2 Mounting MT-3802( )/ARC. - Each Mounting MT-3802( )/ARC shall be packaged Method III as follows: Cushion each mounting by wrapping in paperboard conforming to Type 1 of Specification PPP-P-291. The wrapping shall be secured by tape, Type III conforming to PPP-T-45. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W6c weather resistant, place the technical literature packaged as specified in 5.1.1.4.1 in the box directly under the lid of the box, box closure shall be as specified in the appendix of the box specification.

5.1.1.4.3 Intermediate packaging. - Unit packages, prepared as above, shall be placed in intermediate containers conforming to PPP-B-636, W5c weather resistant. Such containers shall be uniform in size and shapes, shall have minimum cube and tare, shall contain multiples of five units, and shall not exceed ten pounds each. Such containers shall be closed as provided in the appendix to the box specification.

5.1.2 Level C. - Each Mounting MT-3802( )/ARC shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be Level A, B or C as specified. (See 6.2b)

5.2.1 Level A. -

5.2.1.1 Palletized load. - A quantity of items packaged as specified in 5.1.1.4.3 shall be placed on a pallet, Load Type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accomodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather resistant W5c. The load shall be "banded" to the pallet by strapping.

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5.2.1.3 Less than palletized load.- When quantities per destination or less than a pallet load, a quantity of items packaged as specified in 5.1.1.4.3 shall be packed within a close-fitting fiberboard by conforming to PPP-B-636, weather-resistant. Closure shall be in accordance with the appendix of the box specification. Reinforcing shall be by nonmetallic strapping or pressing sensitive filament tape banding conforming to PPP-S-760, Type II or PPP-T-97, Type IV respectfully, selection of the material and application shall be in accordance with the appendix of the box specification.

5.2.2 Level B. - A quantity of Mountings MT-3802( )/ARC as specified in 5.1 shall be packed as specified in 5.2.1.

5.2.3 Level C. - A quantity of Mountings MT-3802( )/ARC packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that pallets, fiberboard boxes and fiberboard caps shall be of the type, size, commonly used for the purpose and shall comply with the rules and regulations of the command carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the packages and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Mounting MT-3802( )/ARC is intended to be used to provide vibration isolation and mounting facilities for security equipment when installed in fixed or rotary wing aircraft.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Levels of preservation, packaging, and packing required. (Level A, Level B, or Level C).

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. Preproduction inspection:

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(1) Three (3) sample units of each item cited in Section 1 are generally required so that lengthly environmental tests can be completed on specified sample units while complete performance measurements can be made on one sample unit. (See 3.3)

(2) Preproduction pack(s) as follows:

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon, (including rough handling).

e. Marking and shipping of sample.

f. Place of final inspection.

g. Technical literature required.

h. Quantity of spare parts required.

i. Submission of the statement of treatment referenced in 3.9 should be sixty (6) days after award of contract. This statement should be submitted to the contracting officer.

6.3 Group C inspection. - Approval to ship unit(s) on order may be withheld, at the discretion of the Government pending decision from the procuring activity, until compliance with 4.5.4.1 is complete.

6.4 Location of operational inspection. - it is desirable that the operational inspection 4.12 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected immediately prior to packaging.

6.5 Nomenclature. - The parentheses in the nomenclautre will be deleted or replaced by a letter identifying the particular design; for example: MT-3802W/ARC. The contractor should apply for noemnclature in accordance with the applicable clause in the contract. (See 1.1)

6.6 Definition of Level B. - Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

6.7 Dimensional data. - Dimensions of packaging materials in the Bill of Materials Table are based on the dimensions of the equipment cited in the table. When dimensions of the items vary, the dimensions of the packaging materials shall be adjusted accordingly.

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6.8 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification by the Government will be adjusted to make maximum utilization of the contractors quality control system and the quality history of the product.

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Preparing activity:
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